PURPOSEPATH JOURNEY MAP

Weather-Based Prediction of Wind Turbine Energy Output:

A Next-Generation Approach to Renewable Energy Management

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SCENARIO: A Wind Farm Operator uses the AI-based system to monitor weather conditions, predict wind turbine energy output, analyze results, and optimize grid stability.

	Entice	>> Enter	Enter	Engage	Extend
Steps	• Operator needs better renewable energy forecasting • Grid instability due to fluctuating wind • Rising demand for renewable energy efficiency. • Organization explores AI-based renewable tools	• Open web dashboard • Enter city / weather location • Fetch real-time weather data	• Open web dashboard • Enter city / weather location • Fetch real-time weather data • Rech meos-inteerface • Backend processing	• System sends weather data to ML model • AI predicts wind turbine energy output • Dashboard displays gauge & analytics charts	• System stores prediction history • Personalized recommendations • Continuous performance optimizat
Interactions	• Researching renewable energy analytics platforms • Internal meetings with grid management team	• React web interface • Weather API integration • Backend processing	• Flask web interface • Machine Learning model(.pkl) • Data visualization (charts & gauges)	• Download report option • Data export feature • Team discussion	• Historical analytics • Performance tracking • Predictive improvements
Goals & motivations	• Help me reduce power imbalance • Seeing potential automation benefits	• Difficulty finding reliable predictive tools	• Help me quickly access weather data • Help me avoid manual data collection	• Clean dashboard interface • Fast API response	- Improved grid stability • Better renewable performance
Positive moments	• Discovering AI-powered forecasting	• Clean dashboard interface	• Help me quickly access weather data	• Prediction delays • Model uncertainty • Low confidence predictions	• Add automated reporting • Add anomaly detection setion
Areas of opportunity	• Provide real-time demo dashboards • Show predictive accuracy reports	• API errors incernect City input	• Auta-detect location • Input valisation & suggestions	• Dewnload epert option • Data export feature • Team discussion	• Adolieng term forecasting module • Add anomaly detection • Cloud scalability